

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics Practical

Practical – XII

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Practical 12: Analysis of Variance and Experimental Design

Part A: Analysis of Variance (ANOVA)

1. Load the Retail and warehouse Sale.csv dataset using Pandas.
2. Display the first five records and identify the number of rows and columns.
3. Display the column names and data types.
4. Check for missing values and duplicate records.
5. Display the different categories present in the ITEM TYPE column.
6. Calculate the average RETAIL SALES for each ITEM TYPE.
7. Create a box plot to compare RETAIL SALES across different ITEM TYPE categories.
8. Formulate the null and alternative hypotheses to determine whether the average retail sales differ among item types.
9. Perform one-way ANOVA using ITEM TYPE as the factor and RETAIL SALES as the response variable.
10. Calculate and display the F-statistic and p-value.
11. Compare the p-value with the significance level of 0.05.
12. State whether the null hypothesis is rejected or not rejected.
13. Write the conclusion about the difference in retail sales among different item types.

CODE:

```
# Practical 12: Analysis of Variance (ANOVA)
# Part A: One-Way ANOVA
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import f_oneway
# -----
# 1. Load Dataset
# -----
file_name = "Retail and warehouse Sale.csv"
df = pd.read_csv(file_name)
# -----
# 2. Display First Five Records
# -----
print("\nFIRST FIVE RECORDS")
print("-----")
print(df.head())
print("\nNumber of Rows :", df.shape[0])
print("Number of Columns :", df.shape[1])
```

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# -----
# 3. Display Column Names and Data Types
# -----
print("\nCOLUMN NAMES")
print("-----")
print(df.columns.tolist())
print("\nDATA TYPES")
print("-----")
print(df.dtypes)
# -----
# 4. Check Missing Values and Duplicate Records
# -----
print("\nMISSING VALUES")
print("-----")
print(df.isnull().sum())
print("\nDUPLICATE RECORDS")
print("-----")
print(df.duplicated().sum())
# -----
# Handle Missing Values in RETAIL SALES
# -----
df["RETAIL SALES"] = df["RETAIL SALES"].fillna(
    df["RETAIL SALES"].median()
)
# -----
# 5. Display Different ITEM TYPE Categories
# -----
print("\nITEM TYPE CATEGORIES")
print("-----")
print(df["ITEM TYPE"].unique())
# -----
# 6. Average RETAIL SALES for Each ITEM TYPE
# -----
print("\nAVERAGE RETAIL SALES BY ITEM TYPE")
print("-----")
average_sales = (
    df.groupby("ITEM TYPE")["RETAIL SALES"]
    .mean()
    .sort_values(ascending=False)
)
print(average_sales)
# -----
# 7. Create Box Plot
# -----
plt.figure(figsize=(10, 6))
sns.boxplot(
    data=df,
    x="ITEM TYPE",
    y="RETAIL SALES"
)

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)
plt.title("Retail Sales by Item Type")
plt.xlabel("Item Type")
plt.ylabel("Retail Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
# -----
# 8. Formulate Hypotheses
# -----
print("\nHYPOTHESES")
print("-----")
print("Null Hypothesis (H0):")
print(
    "The average retail sales are equal "
    "among all ITEM TYPE categories."
)
print("\nAlternative Hypothesis (H1):")
print(
    "At least one ITEM TYPE has a different "
    "average retail sales."
)
# -----
# 9. Perform One-Way ANOVA
# -----
groups = []
for item_type in df["ITEM TYPE"].unique():
    group = df[
        df["ITEM TYPE"] == item_type
    ]["RETAIL SALES"]
    groups.append(group)
f_statistic, p_value = f_oneway(*groups)
# -----
# 10. Display F-Statistic and P-Value
# -----
print("\nONE-WAY ANOVA RESULTS")
print("-----")
print("F-Statistic :", f_statistic)
print("P-Value   :", p_value)
# -----
# 11 & 12. Compare with Significance Level
# -----
alpha = 0.05
print("\nSIGNIFICANCE TEST")
print("-----")
print("Significance Level :", alpha)
if p_value < alpha:
    print("P-value < 0.05")
    print(

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        "Decision: Reject the Null Hypothesis."
    )
else:
    print("P-value >= 0.05")
    print(
        "Decision: Do Not Reject the Null Hypothesis."
    )
# -----
# 13. Conclusion
# -----
print("\nCONCLUSION")
print("-----")
if p_value < alpha:
    print(
        "There is a statistically significant difference "
        "in average retail sales among the different "
        "ITEM TYPE categories."
    )
else:
    print(
        "There is no statistically significant difference "
        "in average retail sales among the different "
        "ITEM TYPE categories."
    )

```

OUTPUT:

```

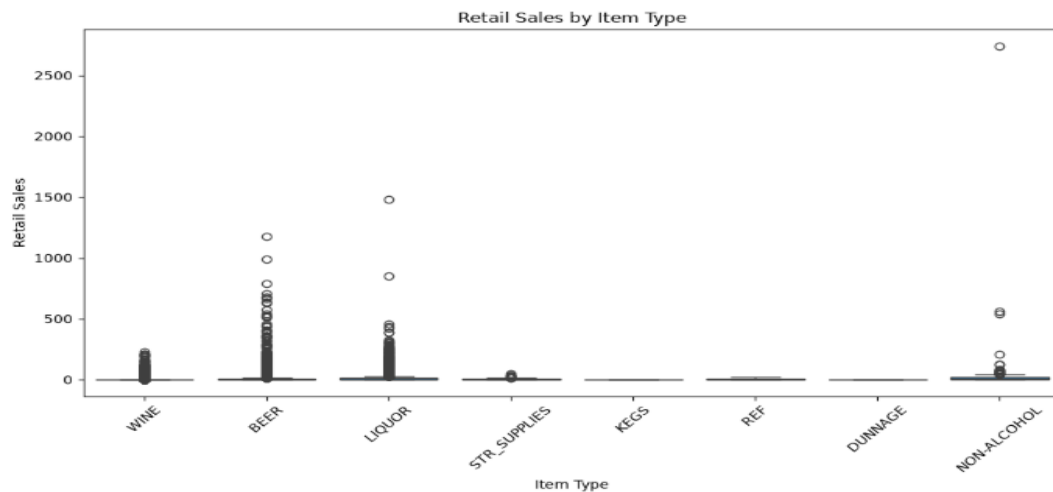
FIRST FIVE RECORDS
-----
   YEAR  MONTH  SUPPLIER ITEM CODE  \
0  2020      1  REPUBLIC NATIONAL DISTRIBUTING CO  100009
1  2020      1                PWSWN INC  100024
2  2020      1          RELIABLE CHURCHILL LLLP  1001
3  2020      1        LANTERNA DISTRIBUTORS INC  100145
4  2020      1        DIONYSOS IMPORTS INC  100293

   ITEM DESCRIPTION  ITEM TYPE  RETAIL SALES  \
0  BOOTLEG RED - 750ML  WINE  0.00
1  MOMENT DE PLAISIR - 750ML  WINE  0.00
2  S SMITH ORGANIC PEAR CIDER - 18.7OZ  BEER  0.00
3  SCHLINK HAUS KABINETT - 750ML  WINE  0.00
4  SANTORINI GAVALA WHITE - 750ML  WINE  0.82

RETAIL TRANSFERS  WAREHOUSE SALES
0  0.0  2.0
1  1.0  4.0
2  0.0  1.0
3  0.0  1.0
4  0.0  0.0

Number of Rows    : 30000
...
WINE              3.195334
KEGS              0.000000
DUNNAGE           0.000000
Name: RETAIL SALES, dtype: float64
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...

```



HYPOTHESES

Null Hypothesis (H_0):

The average retail sales are equal among all ITEM TYPE categories.

Alternative Hypothesis (H_1):

At least one ITEM TYPE has a different average retail sales.

ONE-WAY ANOVA RESULTS

F-Statistic : 123.39469052328639

P-Value : 1.3337307137369178e-179

SIGNIFICANCE TEST

Significance Level : 0.05

P-value < 0.05

Decision: Reject the Null Hypothesis.

CONCLUSION

There is a statistically significant difference in average retail sales among the different ITEM TYPE categories.

Part B: Factorial Experiment

14. Select ITEM TYPE and YEAR as two factors and RETAIL SALES as the response variable.
15. Display the average retail sales for each combination of ITEM TYPE and YEAR.
16. Create a suitable visualization to compare retail sales for different item types across years.
17. Analyze the main effect of ITEM TYPE on retail sales.
18. Analyze the main effect of YEAR on retail sales.
19. Analyze the interaction effect between ITEM TYPE and YEAR.
20. Perform a suitable factorial ANOVA using the selected factors.
21. Display the statistical results and p-values for the main and interaction effects.
22. Identify whether ITEM TYPE, YEAR, or their interaction has a significant effect on RETAIL SALES.
23. Write the final business conclusion based on the ANOVA and factorial experiment results.

CODE:

```
# Practical 12: Factorial Experiment
# Part B: Factorial ANOVA
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```

import statsmodels.api as sm
from statsmodels.formula.api import ols
# -----
# 1. Load Dataset
# -----
file_name = "Retail and warehouse Sale.csv"
df = pd.read_csv(file_name)
# -----
# Handle Missing RETAIL SALES
# -----
df["RETAIL SALES"] = df["RETAIL SALES"].fillna(
    df["RETAIL SALES"].median()
)
# -----
# 14. Select ITEM TYPE and YEAR as Factors
# -----
print("\nFACTORIAL EXPERIMENT")
print("=====")
print("\nFactor 1 : ITEM TYPE")
print("Factor 2 : YEAR")
print("Response : RETAIL SALES")
# -----
# Check YEAR Values
# -----
print("\nYEAR VALUES")
print("-----")
print(df["YEAR"].unique())
print(
    "\nNumber of different years:",
    df["YEAR"].nunique()
)
# -----
# 15. Average RETAIL SALES for Each Combination
# -----
print("\nAVERAGE RETAIL SALES")
print("ITEM TYPE AND YEAR")
print("=====")
average_sales = (
    df.groupby(
        ["ITEM TYPE", "YEAR"]
    )["RETAIL SALES"]
    .mean()
    .reset_index()
)
print(average_sales)
# -----
# 16. Visualization
# -----
plt.figure(figsize=(12, 6))

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sns.barplot(
    data=df,
    x="ITEM TYPE",
    y="RETAIL SALES",
    hue="YEAR",
    errorbar=None
)
plt.title(
    "Retail Sales by Item Type and Year"
)
plt.xlabel("Item Type")
plt.ylabel("Average Retail Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
# -----
# 17. Main Effect of ITEM TYPE
# -----
print("\nMAIN EFFECT OF ITEM TYPE")
print("=====")
item_type_effect = (
    df.groupby("ITEM TYPE")["RETAIL SALES"]
    .mean()
)
print(item_type_effect)
# -----
# 18. Main Effect of YEAR
# -----
print("\nMAIN EFFECT OF YEAR")
print("=====")
year_effect = (
    df.groupby("YEAR")["RETAIL SALES"]
    .mean()
)
print(year_effect)
# -----
#19 & 20. Factorial ANOVA
# -----
print("\nFACTORIAL ANOVA")
print("=====")
# Check whether YEAR has at least two levels
if df["YEAR"].nunique() < 2:
    print("\nWARNING")
    print("-----")
    print(
        "The YEAR column contains only one unique value:"
    )
    print(df["YEAR"].unique())
    print("\nTherefore, a two-factor ANOVA cannot be performed.")

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print("\nReason:")
print(
    "YEAR must contain at least two different levels "
    "to calculate its main effect and interaction."
)
print("\nCannot statistically calculate:")
print("1. Main effect of YEAR")
print("2. ITEM TYPE × YEAR interaction")
print(
    "\nThe ITEM TYPE effect has already been tested "
    "using one-way ANOVA in Part A."
)
else:
    # -----
    # Perform Factorial ANOVA
    # -----
    model = ols(
        "Q('RETAIL SALES') ~ "
        "C(Q('ITEM TYPE')) * C(Q('YEAR'))",
        data=df
    ).fit()
    anova_table = sm.stats.anova_lm(
        model,
        typ=2
    )
    # -----
    # 21. Display Statistical Results
    # -----
    print("\nANOVA TABLE")
    print("-----")
    print(anova_table)
    print("\nP-VALUES")
    print("-----")
    print(anova_table["PR(>F)"])
    # -----
    # 22. Identify Significant Effects
    # -----
    print("\nSIGNIFICANT EFFECTS")
    print("-----")
    for effect, p_value in anova_table["PR(>F)"].items():
        if pd.notna(p_value):
            if p_value < 0.05:
                print(
                    effect,
                    ": Significant"
                )
            else:
                print(
                    effect,

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```

        ": Not Significant"
    )
# -----
# 23. Final Business Conclusion
# -----
print("\nFINAL BUSINESS CONCLUSION")
print("=====")
if df["YEAR"].nunique() < 2:
    print(
        "The uploaded dataset contains only one year, 2020."
    )
    print(
        "Therefore, a complete factorial experiment using "
        "ITEM TYPE and YEAR cannot be performed."
    )
    print(
        "A dataset containing at least two different years "
        "is required to test the YEAR effect and the "
        "ITEM TYPE × YEAR interaction."
    )
else:
    print(
        "The factorial ANOVA results can be used to determine "
        "the significant effects of ITEM TYPE, YEAR, and "
        "their interaction."
    )

```

OUTPUT:

```

---
      FACTORIAL EXPERIMENT
      =====

      Factor 1 : ITEM TYPE
      Factor 2 : YEAR
      Response : RETAIL SALES

      YEAR VALUES
      -----
      [2020]

      Number of different years: 1

      AVERAGE RETAIL SALES
      ITEM TYPE AND YEAR
      =====

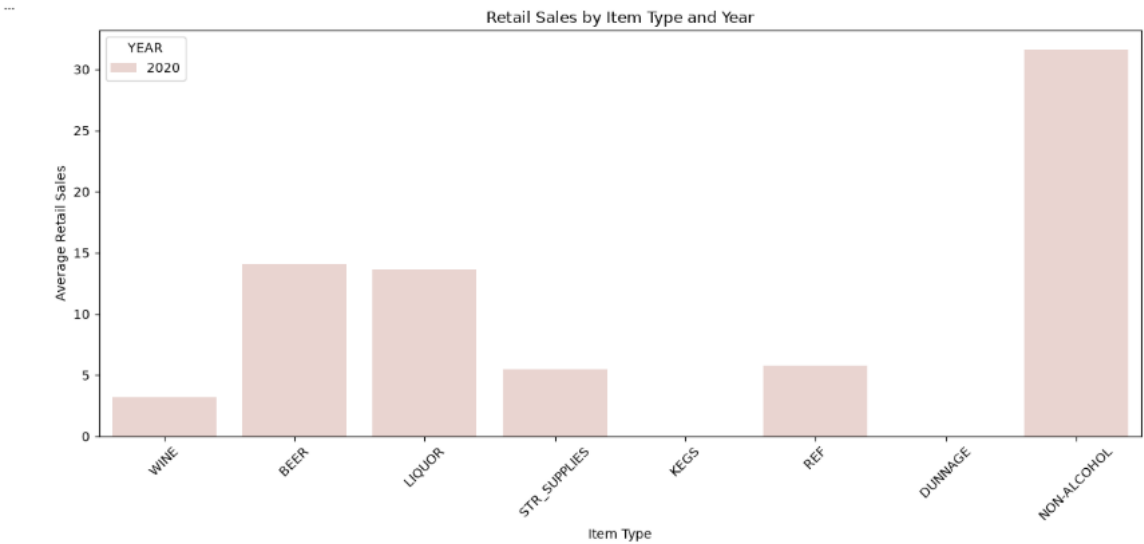
```

	ITEM TYPE	YEAR	RETAIL SALES
0	BEER	2020	14.118748
1	DUNNAGE	2020	0.000000
2	KEGS	2020	0.000000
3	LIQUOR	2020	13.635171
4	NON-ALCOHOL	2020	31.596204
5	REF	2020	5.783750
6	STR_SUPPLIES	2020	5.485714
7	WINE	2020	3.195334

```

---

```



```
...
MAIN EFFECT OF ITEM TYPE
=====
ITEM TYPE
BEER      14.118748
DUNNAGE    0.000000
KEGS       0.000000
LIQUOR     13.635171
NON-ALCOHOL 31.596204
REF        5.783750
STR_SUPPLIES 5.485714
WINE       3.195334
Name: RETAIL SALES, dtype: float64

MAIN EFFECT OF YEAR
=====
YEAR
2020      6.93957
Name: RETAIL SALES, dtype: float64

FACTORIAL ANOVA
=====

WARNING
-----
...
=====
The uploaded dataset contains only one year, 2020.
Therefore, a complete factorial experiment using ITEM TYPE and YEAR cannot be performed.
A dataset containing at least two different years is required to test the YEAR effect and the ITEM TYPE x YEAR interaction.
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```